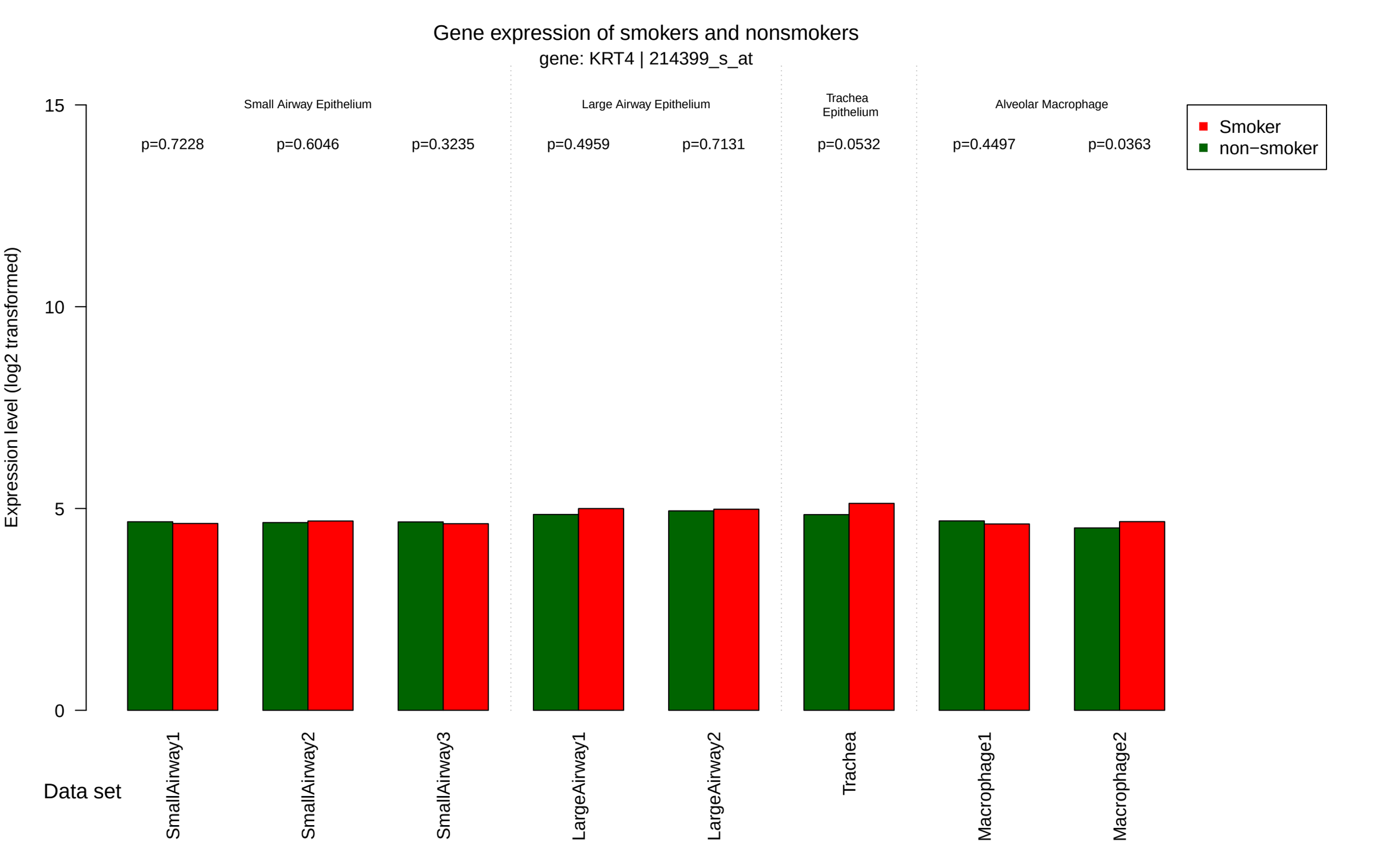
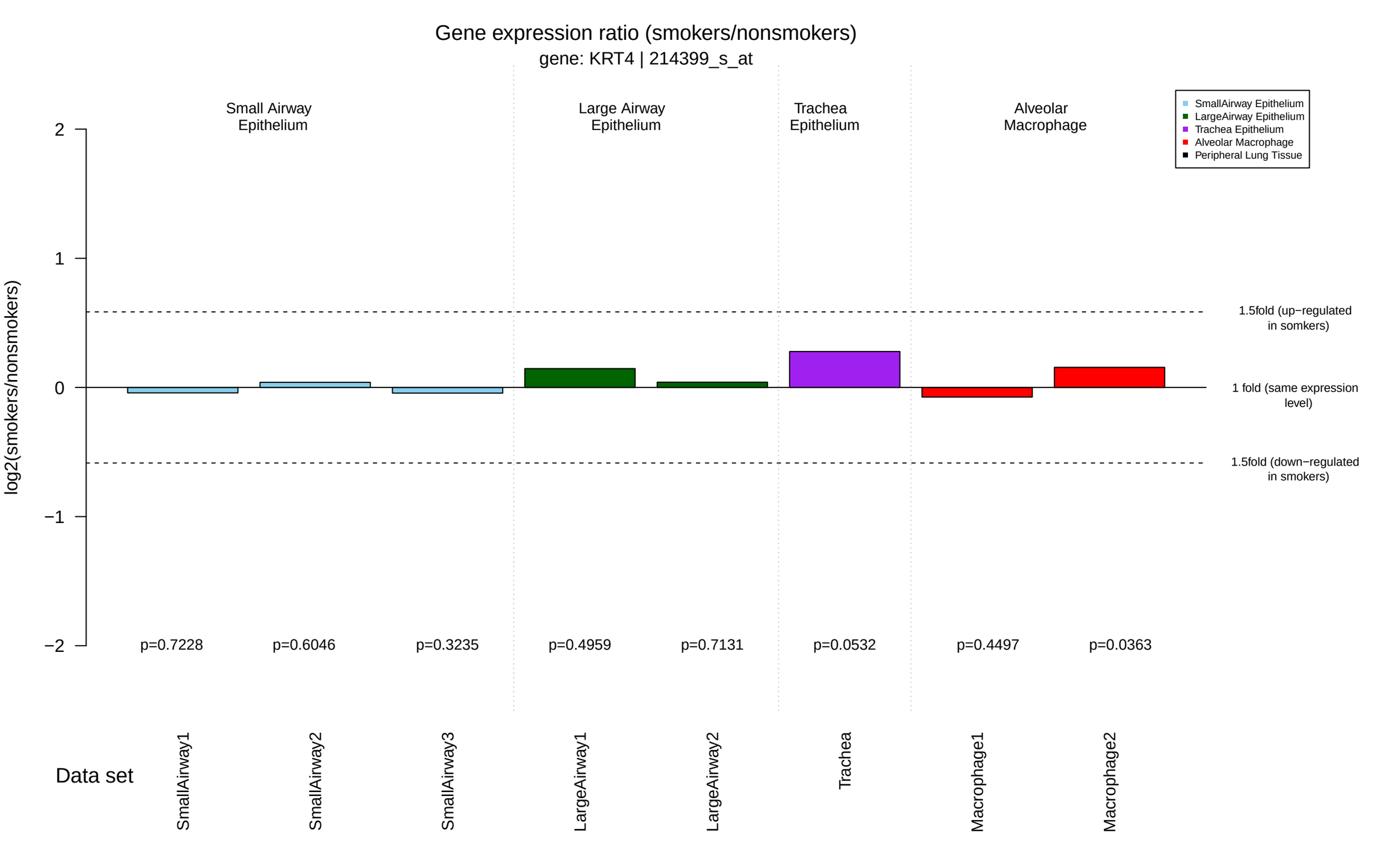
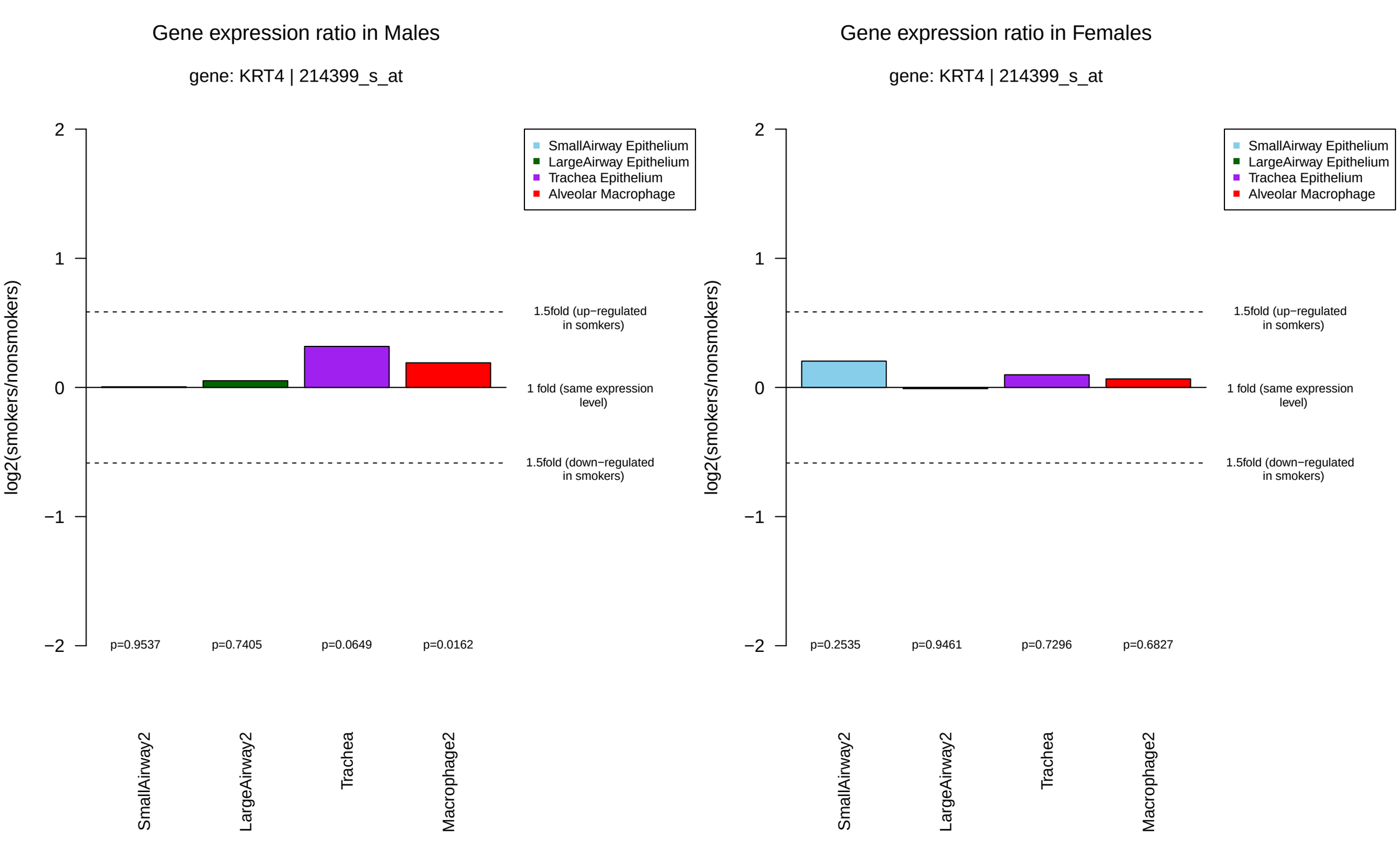
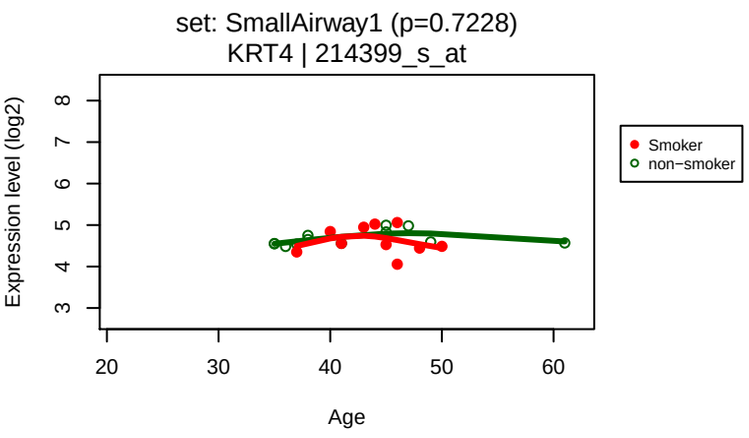
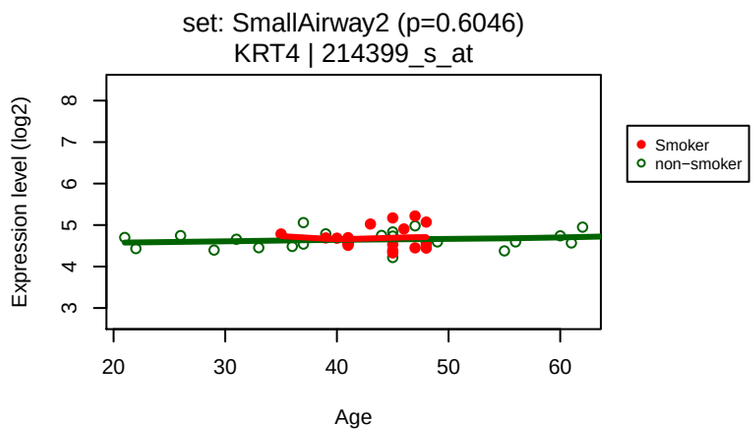
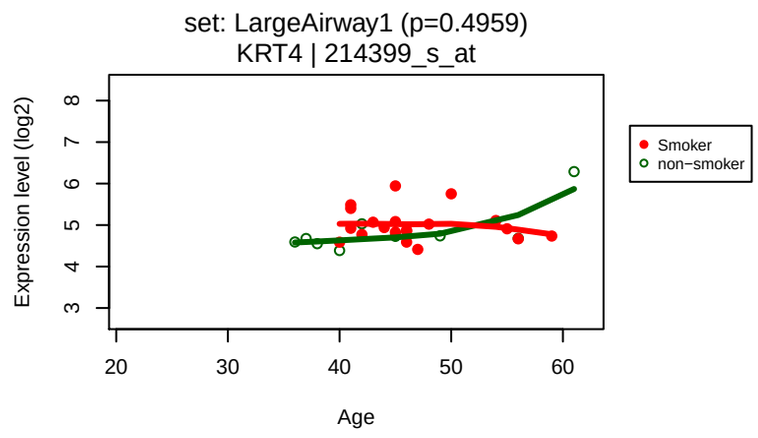

Gene expression of smokers and nonsmokers
gene: KRT4 | 214399_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7228
p=0.6046
p=0.3235
p=0.4959
p=0.7131
p=0.0532
p=0.4497
p=0.0363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

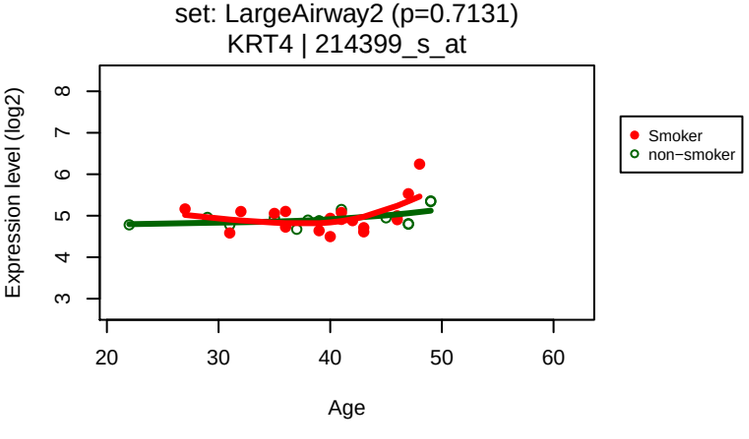

Gene expression ratio (smokers/nonsmokers)
gene: KRT4 | 214399_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7228
p=0.6046
p=0.3235
p=0.4959
p=0.7131
p=0.0532
p=0.4497
p=0.0363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

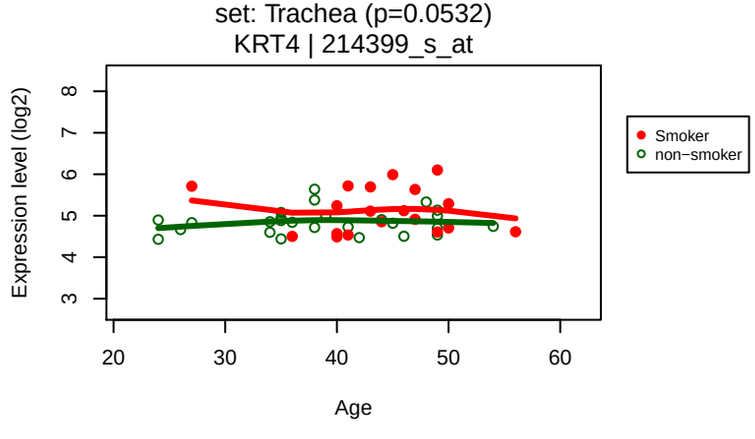

Gene expression ratio in Males
gene: KRT4 | 214399_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9537
p=0.7405
p=0.0649
p=0.0162
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRT4 | 214399_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2535
p=0.9461
p=0.7296
p=0.6827
SmallAirway2
LargeAirway2
Trachea
Macrophage2

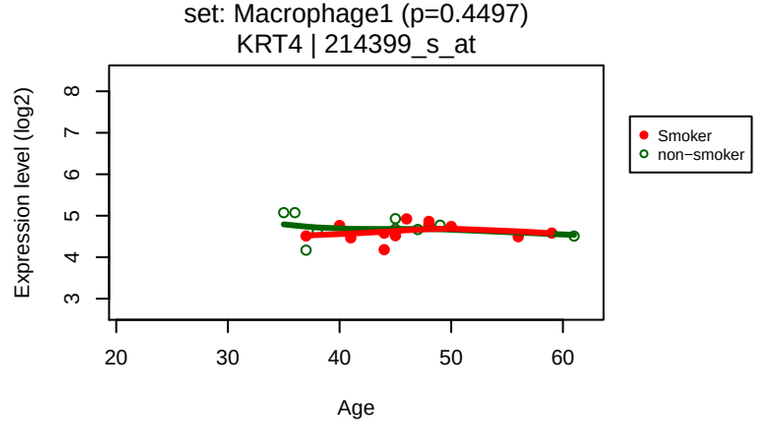

set: SmallAirway1 (p=0.7228)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker

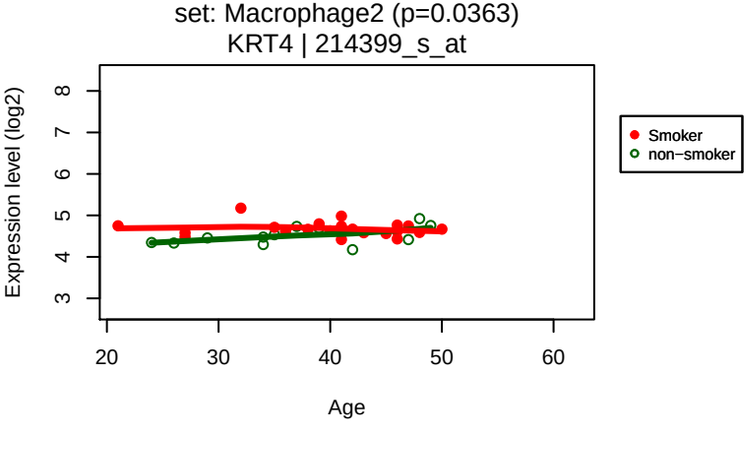

set: SmallAirway2 (p=0.6046)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker

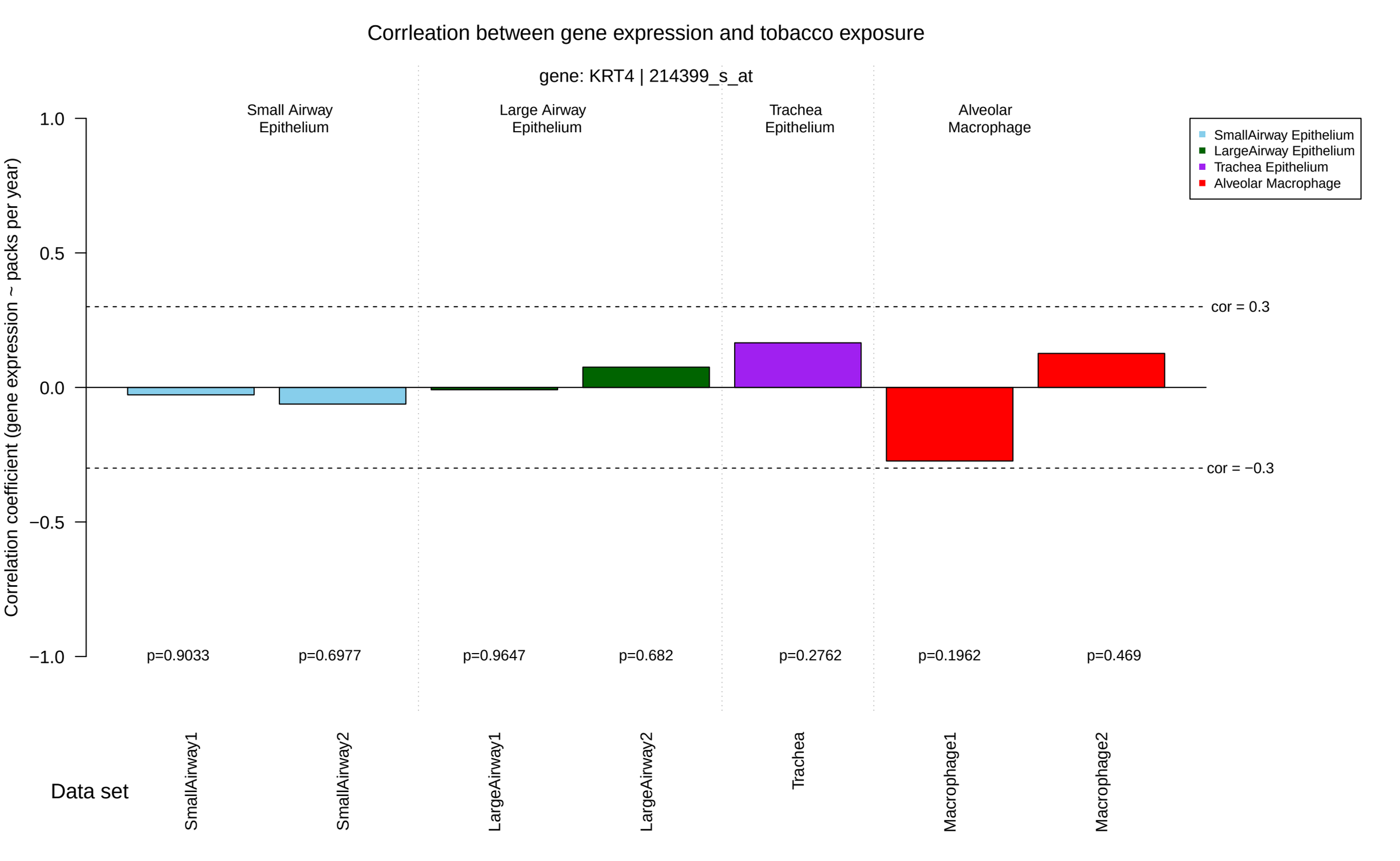

set: LargeAirway1 (p=0.4959)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7131)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0532)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4497)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0363)
KRT4 | 214399_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRT4 | 214399_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9033
p=0.6977
p=0.9647
p=0.682
p=0.2762
p=0.1962
p=0.469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2